

An Internship Report
Of
DecodeLabs AI-ML Virtual Internship

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By
Roll Number of Student: **25SCS1003000858**
Name of Student: **VANSH**
Batch: **2025-2029**

CANDIDATE'S DECLARATION

I, VANSI, do hereby declare that the internship report titled “DecodeLabs AI-ML Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: **VANSI**

Roll No.: **25SCS1003000858**

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the DecodeLabs Virtual Internship Program team for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Artificial Intelligence and Machine Learning. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: **VANSH**

Roll No.: **25SCS1003000858**

TABLE OF CONTENTS

1. Candidate's Declaration ...	i
2. Acknowledgement ...	ii
3. Internship Completion Certificate ...	–
4. Project Description ...	1
4.1 Introduction ...	1
4.2 Organization Profile ...	2
4.3 Problem Statement ...	3
4.4 Project Objectives ...	3
4.5 Scope of the Project ...	4
4.6 Technologies and Tools Used ...	4
4.7 System Architecture ...	5
4.8 Methodology ...	6
4.9 Expected Outcomes ...	7
4.10 Certificates of Completion and Communication Proof ...	8
5. Bibliography/References ...	9

3. INTERNSHIP COMPLETION CERTIFICATE



DECODELABS
VIRTUAL INTERNSHIP PROGRAM

STUDENT ID
AI086328

GLOBAL INTERNSHIP CREDENTIAL

CERTIFICATE OF COMPLETION

DECODELABS VIRTUAL INTERNSHIP

THIS CERTIFICATE IS PROUDLY PRESENTED TO

Vansh

VIRTUAL INTERNSHIP — ARTIFICIAL INTELLIGENCE (AI)

Has successfully completed the **DecodeLabs Virtual Internship Program** in **Artificial Intelligence (AI)**, demonstrating dedication, consistency, and hands-on problem-solving skills across real-world projects and collaborative tasks.

We appreciate your hard work and growth mindset throughout the internship and wish you the very best in your future endeavors.

DURATION
July 03, 2026 — August 03, 2026

DATE OF ISSUE
August 04, 2026



EXCELLENCE • INTEGRITY •
GROWTH
Achievement mark

Official mark



SCAN TO VERIFY



DECODELABS — AUTHORIZED SIGNATORY

Internship Offer Letter

Decode Labs

YOUR DIGITAL LAB

+91 92360 11887 / 94739 35188

hr@decodelabs.tech



GOVT. REGISTERED ENTERPRISE

Date: 3 July 2026

To: Vansh

Date: 3 July 2026

OFFER OF INTERNSHIP

Dear Vansh,

Following your application and assessment, we are delighted to offer you an internship position at **Decode Labs**. We believe that your skills and academic background will be a valuable asset to our programs.

The details of your internship offer are as follows:

Position	Artificial Intelligence (AI) Intern
Department / Track	Artificial Intelligence (AI)
Start Date	3 July 2026
End Date	3 August 2026
Mode of Work	Remote / Virtual

During this internship, you will be expected to work on assigned projects, complete milestones on time, and participate in mentor-led sessions. This is a learning-focused internship designed to provide you with hands-on industry experience and professional growth.

Upon successful completion of the internship and required project submissions, you will be awarded a verified Certificate of Completion and a Letter of Recommendation based on your performance.

We welcome you to Decode Labs and look forward to a mutually rewarding learning experience.

Warm regards,

Decode Labs Team

www.decodelabs.tech



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the Virtual Internship undertaken under the DecodeLabs Virtual Internship Program, in the domain of “Artificial Intelligence (AI) / AI-ML”. The internship was carried out from July 2026 to August 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida.

The primary aim of this internship was to build a strong practical foundation in Machine Learning and Artificial Intelligence, and to apply it to real-world problems such as image classification, object detection, and intelligent application development.

The internship followed a structured curriculum combining conceptual learning with hands-on projects, mentor-led sessions, and project submissions. On successful completion of all the requirements, a Certificate of Completion was awarded by DecodeLabs.

4.2 Organization Profile

DecodeLabs is a government-registered digital learning and technology enterprise operating under the theme “Your Digital Lab”. DecodeLabs runs structured Virtual Internship Programs designed to provide engineering students with industry-aligned, hands-on exposure in emerging domains such as Artificial Intelligence and Machine Learning. The programs emphasize real-world projects, collaborative tasks, mentor guidance, and professional growth, helping students build practical skills and a growth mindset.

4.3 Problem Statement

Although a large number of engineering students learn the theoretical foundations of machine learning and neural networks during their academic coursework, very few get structured, hands-on exposure to building and applying AI models using industry-relevant tools and workflows. Concepts such as convolutional neural networks, object detection, and image classification are often understood only at a theoretical level, with little practice in training, evaluating, and integrating such models into real applications. The problem addressed by this internship was to practically learn and apply the AI-ML workflow – from building and training models, to detecting and classifying objects, to developing intelligent features for applications.

4.4 Project Objectives

- To understand the first principles of machine learning and build models for image recognition and classification.
- To learn the basics of object detection and integrate detectors into applications.
- To train and evaluate custom models for computer-vision tasks.
- To build intelligent features such as product image search using on-device or cloud-based detection.
- To detect objects in static images and live feeds for visual search applications.
- To understand backend integration concepts for AI-powered mobile or web applications.
- To build custom image-classification models and improve overall classification skill.

- To create and integrate a custom image-classifier model into an application.
- To successfully complete all project milestones and earn the Certificate of Completion.

4.5 Scope of the Project

The scope of this internship spans the complete AI-ML computer-vision workflow – starting from understanding neural-network fundamentals and building classification models, moving to object detection using pretrained and custom-trained models, and extending into product image search features that combine detection with visual search. It further covers building and improving custom image-classification models and integrating them into applications. The internship was limited to a virtual, project-based learning environment provided by DecodeLabs, and did not involve deployment on any live organizational production system.

4.6 Technologies and Tools Used

- Programming & ML Framework: Python, TensorFlow / Keras (or equivalent AI libraries)
- On-device / Efficient ML: Lightweight model formats suitable for mobile or edge deployment
- Computer Vision: Convolutional Neural Networks (CNNs), object detectors, custom image classifiers
- Application Layer: Integration concepts for mobile / web applications, camera-feed based detection
- Learning Platform: DecodeLabs Virtual Internship Program – mentor sessions, projects, and assessments

4.7 System Architecture

The general architecture followed during the internship for the AI-ML learning track can be summarized in the following layers:

- **Model Layer:** A neural network (e.g., convolutional neural network) built and trained forms the core intelligence layer for image recognition and classification.
- **Conversion / Optimization Layer:** Trained models are converted or optimized into lightweight formats suitable for efficient inference.
- **Detection Layer:** Pretrained and custom-trained object detectors identify and localize objects within static images and live camera feeds.
- **Search Layer:** A product image search feature combines object detection with backend integration to enable visual product search.
- **Classification Layer:** A custom image classifier further categorizes detected objects/images with improved accuracy.
- **Application Layer:** The trained and optimized models are integrated into an application, consuming the detection, search, and classification capabilities.
- **Assessment Layer:** Project milestones, mentor feedback, and final evaluation track progress and successful completion.

4.8 Methodology

The internship followed a structured methodology combining conceptual learning, hands-on projects, and mentor-guided assessments, culminating in project submissions and Certificate of Completion, as summarized conceptually below (adapted to the DecodeLabs program duration of approximately four weeks).

Phase	Module / Focus	Description
1	Foundations of ML & Neural Networks	Foundational concepts of machine learning, from first principles through to building models for image recognition and classification.
2	Object Detection Basics	Basics of object detection and integrating pretrained object detectors into applications.
3	Custom Models & Product Image Search	Training custom models and building product image search features using detection on static images and live feeds.
4	Integration & Classification	Backend integration concepts, custom image-classification models, and integrating a classifier into an application.
Final	Project Submission & Evaluation	Final project submissions, mentor evaluation, and award of Certificate of Completion.

Note: There was no stipend associated with this internship.

Each phase included project work and assessments to evaluate conceptual understanding and practical skills. The overall evaluation was based on performance across milestones and final project submissions.

4.9 Expected Outcomes

- Ability to explain and apply the first principles of machine learning to build models for image-related tasks.
- Practical understanding of object detection, including the use of pretrained and custom-trained models.
- Hands-on exposure to building a product image search feature combining detection with application integration.
- Capability to build, evaluate, and improve custom image-classification models.
- Skill to integrate trained models into an application context.
- Successful completion of the DecodeLabs Virtual Internship Program in Artificial Intelligence (AI), resulting in a verified Certificate of Completion.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Certificate of Completion issued by DecodeLabs, confirming successful completion of the Virtual Internship Program in Artificial Intelligence (AI), are attached in Chapter 3 of this report. These documents serve as official proof of the internship undertaken from 3 July 2026 to 3 August 2026.

5. BIBLIOGRAPHY/REFERENCES

1. DecodeLabs, “Offer of Internship – Artificial Intelligence (AI) Intern,” dated 3 July 2026, Student: Vansh.
2. DecodeLabs, “Certificate of Completion – DecodeLabs Virtual Internship Program in Artificial Intelligence (AI),” Student ID: AI086328, Duration: July 03, 2026 — August 03, 2026, Date of Issue: August 04, 2026.
3. TensorFlow Documentation, <https://www.tensorflow.org/>
4. TensorFlow Lite Documentation, <https://www.tensorflow.org/lite>
5. Google Developers – Machine Learning, <https://developers.google.com/machine-learning>
6. DecodeLabs Official Website, <https://www.decode labs.tech>